COURSE 2 · SECTION 4 · WORKED SOLUTION

Late drawings, findings, mass and cost

Section 4 · Drawing release, review findings, mass/cost vs target

DESIGN QUESTION

How often are drawings released on time, what do design reviewers find most often, and which part revisions miss their mass and cost targets?

Steps

1. Open Section05_1_Start.pbix (or Engxtech_S04.pbix) > File > Save as > Engxtech_S04_Exercise.
2. Drawings page: cards On-time %, Late Drawings, Avg Days Late; table DrawingNo, PartRev, ReviewRounds, DaysLate sorted by DaysLate.
3. Findings page: the Pareto is a line and clustered column chart: X axis ReviewFindings[Category]; columns Finding Count; line Findings Cum %; sorted by Finding Count, descending.
4. Right-click _Measures > New measure: Tolerance Share (below). Format Percentage, 1 decimal. Put it on a card.
5. Targets page: cards Revs Over Mass Target and Revs Over Cost Target; two bar charts by MassCost[PartRev] titled Mass vs target (%) and Cost vs target (%).

DAX from the video

```
Drawings[DaysLate] = DATEDIFF ( Drawings[PlannedRelease], Drawings[ActualRelease], DAY )
Drawing Count    = COUNTROWS ( Drawings )
On-time Drawings = CALCULATE ( [Drawing Count], Drawings[DaysLate] <= 0 )
On-time %        = DIVIDE ( [On-time Drawings], [Drawing Count] )
Avg Days Late    = CALCULATE ( AVERAGE ( Drawings[DaysLate] ), Drawings[DaysLate] > 0 )
Finding Count    = COUNTROWS ( ReviewFindings )
Findings Cum % =
VAR cur  = [Finding Count]
VAR cats = ALLSELECTED ( ReviewFindings[Category] )
VAR upto = SUMX ( FILTER ( cats, CALCULATE ( [Finding Count] ) >= cur ), CALCULATE ( [Finding Count] ) )
RETURN DIVIDE ( upto, CALCULATE ( [Finding Count], cats ) )
Mass vs Target % = DIVIDE ( [Actual Mass] - [Target Mass], [Target Mass] )
Revs Over Mass Target =
COUNTROWS ( FILTER ( MassCost, MassCost[ActualMass_g] > RELATED ( Parts[TargetMass_g] ) ) )
```

DAX new in this exercise

```
Tolerance Share =
DIVIDE (
    CALCULATE ( [Finding Count], ReviewFindings[Category] = "Tolerance" ),
    CALCULATE ( [Finding Count], ALL ( ReviewFindings[Category] ) )
)
```

DATEDIFF returns a negative number when the drawing came out early, so on-time uses <= 0. The P300-B drawing was the latest at 19 days after 3 review rounds. Mass and cost tell different stories: almost every revision is heavier than target, while the biggest cost overrun (P400-A) is not the biggest mass overrun (P100-B).

Expected result

Check	Result
On-time release	**33.3 % (7/21)**
Average lateness of the late drawings	**4.6 d**
P300-B drawing: days late / review rounds	**19 d / 3**
Findings / of which major	**60 / 19**
Top Pareto bar	**Tolerance 23**
Tolerance Share	**38.3 %**
Cumulative % after the first three bars	**71.7 %**
Revisions over mass target	**15 of 16**
Revisions over cost target	**10 of 16**
Largest mass / cost overrun	**P100-B +12.4 % / P400-A +21.8 %**